CASE

Doris, 76 · several syncopal episodes · pulse 35 · BP 100/60 · EKG: P waves at 75/min, QRS at 35/min, no relationship between them · **third-degree (complete) AV block**

BIO 004 · WEEK 3 · CASE DEEP DIVE

The Cardiac Conduction System

An older patient who feels dizzy when she stands, a pulse of 35, and an EKG with atria and ventricles beating independently.

PATIENT CASE

Doris is 76. Over the past few weeks she has had **three brief fainting spells**, each lasting seconds, each followed by spontaneous recovery. Today she nearly fell again and her daughter brought her to the ED.

On exam Doris is alert. **Pulse 35**, regular. BP 100/60. The EKG is the diagnosis: **P waves marching along at about 75 per minute, QRS complexes appearing independently at about 35 per minute, no consistent relationship between them**. The atria and ventricles are no longer talking to each other. Doris has **complete (third-degree) AV block**. The cardiology team is being called to place a permanent pacemaker.

The heart's ability to beat as a coordinated pump depends on a specialized network of cells that generate and conduct electrical impulses. When part of that network fails, you can see the result — at the bedside, on the EKG, and in the patient's symptoms.

> GUIDING QUESTION
>
> What cells generate the heartbeat, what path does the impulse take from atrium to ventricle, and how does a single point of failure produce two separate rhythms in one heart?

PREWORK

Companion to your Week 3 Cardiac Conduction prework page. Terms, video, and spaced recall for the SA node, AV node, bundle of His, bundle branches, Purkinje fibers, and the cardiac action potential live there.

1 TWO CELL TYPES

Workers and pacemakers

The heart has two functional populations of cardiac muscle cells. **Contractile cells** (the vast majority) are striated muscle cells that do the actual pumping. **Conducting cells** are specialized cardiac muscle cells that generate and propagate electrical signals; they contract weakly or not at all. The contractile cells are the engine; the conducting cells are the wiring and the spark plugs.

2 THE PATHWAY

From SA node to Purkinje fibers, in order

1. SA (sinoatrial) node

In the right atrial wall near the entry of the SVC. The heart's natural pacemaker. Fires at ~60-100 beats/min. Fed by the RCA in most people.

2. Atrial conduction

Impulse spreads across both atria via gap junctions between contractile cells, depolarizing them and producing the *P wave*. Atria contract.

3. AV (atrioventricular) node

At the floor of the right atrium near the tricuspid valve. The ONLY electrical connection between atria and ventricles (the rest of the AV ring is insulating fibrous tissue). Slows conduction (~0.1 second delay), giving atria time to empty before ventricles contract. Fed by the RCA in ~90% of people.

4. Bundle of His

The AV node's output. Passes through the interventricular septum.

5. Right and left bundle branches

The bundle of His splits into right and left branches running down each side of the interventricular septum.

6. Purkinje fibers

Fan out through the ventricular myocardium. Fast conduction delivers the impulse to the entire ventricular mass nearly simultaneously, producing the *QRS complex*. Ventricles contract.

Why the AV node's delay matters. Without the 0.1-second AV delay, atria and ventricles would contract simultaneously, leaving the ventricles only partially filled. The AV node's slow conduction is exactly what makes a coordinated, efficient heartbeat possible.

3 PACEMAKER HIERARCHY

Backup generators with progressively slower rates

The SA node is normally the fastest pacemaker and sets the heart's rhythm. But if the SA node fails or its signal can't reach downstream tissue, other parts of the conduction system can take over — at progressively slower intrinsic rates:

PACEMAKER	INTRINSIC RATE	TAKES OVER IF...
SA node	60-100/min	(normal)
AV node / junctional	40-60/min	SA fails OR atrial signal can't reach it (junctional escape rhythm)
Bundle of His / Purkinje (ventricular)	20-40/min	AV node fails (idioventricular escape rhythm — what Doris has)

Doris's ventricles are beating at 35/min on their own because the impulse from her atria can no longer cross the AV node. The atrial pacemaker keeps firing at 75; her ventricles fall back to their own intrinsic ventricular pacemaker.

What each wave shows

The EKG records the heart's electrical activity from the body surface. Each beat generates a recognizable pattern:

P wave
Atrial depolarization (signal spreading from SA node across the atria).

PR interval
Time from start of atrial depolarization to start of ventricular depolarization. Mostly the AV node's delay. Normal 120-200 ms.

QRS complex
Ventricular depolarization. Brief and narrow because Purkinje conduction is fast.

T wave
Ventricular repolarization (resetting for the next beat).

(Atrial repolarization)
Buried inside the QRS, not usually visible.

5 AV BLOCK GRADES

Three flavors of AV node failure

DEGREE	WHAT'S HAPPENING	EKG FINDING
1st	AV node conducts every P to a QRS but slowly	PR interval prolonged > 200 ms; every P has a QRS
2nd, Mobitz I (Wenckebach)	AV node progressively tires, eventually drops a beat	PR interval gradually lengthens until a P wave is not followed by a QRS
2nd, Mobitz II	AV node intermittently blocks without warning	Constant PR interval; occasional P waves with no QRS
3rd (complete)	NO atrial impulses reach the ventricles; ventricles escape on their own	P waves and QRS complexes march independently. *Doris.*

6 AUTONOMIC CONTROL

How the body adjusts heart rate

The SA node's intrinsic rate is ~100/min. Resting heart rate is lower (60-80) because **parasympathetic (vagal) tone** dominates at rest, slowing the SA node. With exercise, vagal tone withdraws and **sympathetic input** increases, speeding the SA node and the AV node, increasing contractility, and shortening conduction times.

Both autonomic limbs reach the SA and AV nodes; sympathetic also reaches the ventricular myocardium. This is why beta-blockers (which block sympathetic effects) slow the heart and weaken contractility, and why atropine (which blocks parasympathetic effects) speeds it.

✎ **DRAW ZONE 1** -- *Sketch the heart with the conduction system overlaid. Label SA node, AV node, bundle of His, bundle branches, Purkinje fibers. Then mark where complete block can occur.*

The block can be at the AV node itself, the bundle of His, or below in both bundle branches. Color each potential site.

✎ **DRAW ZONE 2** *-- Draw an EKG rhythm strip for: normal sinus rhythm, 1st degree AV block, Mobitz I, Mobitz II, and 3rd degree (Doris). Label P waves and QRS complexes.*

For 3rd degree, draw the P waves at one rate, QRS at a slower rate, and show them marching independently.

RETURNING TO DORIS

Why she's syncoping and why she needs a pacemaker

Complete heart block is a mechanical and electrical problem. Doris's ventricles are firing on their own at 35/min — too slow for her brain's perfusion needs, especially during postural change.

"*Pulse 35, EKG dissociation*"

AV node has stopped conducting atrial impulses to ventricles. Atria fire at 75 (SA node). Ventricles escape on their own intrinsic ventricular pacemaker at ~35.

"*Syncope on standing*"

Ventricular rate of 35 cannot increase to meet the demand for cerebral perfusion when she stands. BP drops, cerebral perfusion drops, she briefly loses consciousness.

"*Why now? What caused the block?*"

Most common causes in this age group: degenerative fibrosis of the conduction system (Lev/Lenègre disease), ischemia (especially RCA territory affecting the AV node), medications (beta-blockers, calcium-channel blockers, digoxin), Lyme disease in endemic areas.

"*Risk of cardiac arrest*"

The ventricular escape rhythm at 35 is unstable. If it slows further or stops (asystole), she dies. This is why she needs a permanent pacemaker urgently.

"*Temporary vs permanent pacing*"

Transcutaneous (external pads) or transvenous temporary pacing buys time. A permanent pacemaker (small generator + leads to the right atrium and/or right ventricle) is the definitive treatment.

"*Will a pacemaker fix it?*"

Yes, completely. The pacemaker senses absent ventricular activity and fires the ventricle at a normal rate. Modern dual-chamber pacemakers can also pace the atrium and synchronize atrial and ventricular contraction (DDD mode), restoring nearly normal hemodynamics.

SA, AV, His, Purkinje. Doris's EKG is the conduction system telling you exactly where the wire broke. When you go back to the prework, the conduction diagram will land differently — it's now the schematic the pacemaker is replacing.